Fourier Analysis

Exercise

Contents

Fourier Analysis: An Introduction **1**
Chapter 1. The Genesis of Fourier Analysis 1
Chapter 2. Basic Properties of Fourier Series 12
Chapter 3. Convergence of Fourier Series 32
Chapter 4. Some Applications of Fourier Series 52
Chapter 5. The Fourier Transform on $\mathbf{R}$ 65
Chapter 6. The Fourier Transform on $\mathbf{R}^d$ 88
Chapter 7. Finite Fourier Analysis 103
Chapter 8. Dirichlet's Theorem 116

Fourier Analysis: An Introduction

Chapter 1. The Genesis of Fourier Analysis

1. If $z = x + iy$ is a complex number with $x, y \in \mathbb{R}$, we define

$$|z| = (x^2 + y^2)^{1/2}$$

and call this quantity the **modulus** or **absolute value** of z.

(a) What is the geometric interpretation of $|z|$?

(b) Show that if $|z| = 0$, then $z = 0$.

(c) Show that if $\lambda \in \mathbb{R}$, then $|\lambda z| = |\lambda||z|$, where $|\lambda|$ denotes the standard absolute value of a real number.

(d) If z_1 and z_2 are two complex numbers, prove that

$$|z_1 z_2| = |z_1||z_2| \qquad \text{and} \qquad |z_1 + z_2| \leq |z_1| + |z_2|.$$

(e) Show that if $z \neq 0$, then $|1/z| = 1/|z|$.

2. If $z = x + iy$ is a complex number with $x, y \in \mathbb{R}$, we define the **complex conjugate** of z by

$$\overline{z} = x - iy.$$

(a) What is the geometric interpretation of $\overline{z}$?

(b) Show that $|z|^2 = z\overline{z}$.

(c) Prove that if z belongs to the unit circle, then $1/z = \overline{z}$.

3. A sequence of complex numbers $\{w_n\}_{n=1}^{\infty}$ is said to converge if there exists $w \in \mathbb{C}$ such that

$$\lim_{n\to\infty} |w_n - w| = 0,$$

and we say that w is a limit of the sequence.

(a) Show that a converging sequence of complex numbers has a unique limit.

The sequence $\{w_n\}_{n=1}^{\infty}$ is said to be a **Cauchy sequence** if for every $\epsilon > 0$ there exists a positive integer N such that

$$|w_n - w_m| < \epsilon \qquad \text{whenever } n, m > N.$$

(b) Prove that a sequence of complex numbers converges if and only if it is a Cauchy sequence. [Hint: A similar theorem exists for the convergence of a sequence of real numbers. Why does it carry over to sequences of complex numbers?]

A series $\sum_{n=1}^{\infty} z_n$ of complex numbers is said to converge if the sequence formed by the partial sums

$$S_N = \sum_{n=1}^{N} z_n$$

converges. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of non-negative real numbers such that the series $\sum_n a_n$ converges.

(c) Show that if $\{z_n\}_{n=1}^{\infty}$ is a sequence of complex numbers satisfying $|z_n| \leq a_n$ for all n, then the series $\sum_n z_n$ converges. [Hint: Use the Cauchy criterion.]

4. For $z \in \mathbb{C}$, we define the **complex exponential** by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

(a) Prove that the above definition makes sense, by showing that the series converges for every complex number z. Moreover, show that the convergence is uniform[5] on every bounded subset of $\mathbb{C}$.

(b) If z_1, z_2 are two complex numbers, prove that $e^{z_1} e^{z_2} = e^{z_1+z_2}$. [Hint: Use the binomial theorem to expand $(z_1 + z_2)^n$, as well as the formula for the binomial coefficients.]

(c) Show that if z is purely imaginary, that is, $z = iy$ with $y \in \mathbb{R}$, then

$$e^{iy} = \cos y + i \sin y.$$

This is Euler's identity. [Hint: Use power series.]

(d) More generally,

$$e^{x+iy} = e^x(\cos y + i \sin y)$$

whenever $x, y \in \mathbb{R}$, and show that

$$|e^{x+iy}| = e^x.$$

(e) Prove that $e^z = 1$ if and only if $z = 2\pi k i$ for some integer k.

(f) Show that every complex number $z = x + iy$ can be written in the form

$$z = re^{i\theta},$$

where r is unique and in the range $0 \le r < \infty$, and $\theta \in \mathbb{R}$ is unique up to an integer multiple of 2π. Check that

$$r = |z| \quad \text{and} \quad \theta = \arctan(y/x)$$

whenever these formulas make sense.

(g) In particular, $i = e^{i\pi/2}$. What is the geometric meaning of multiplying a complex number by i? Or by $e^{i\theta}$ for any $\theta \in \mathbb{R}$?

(h) Given $\theta \in \mathbb{R}$, show that

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These are also called Euler's identities.

(i) Use the complex exponential to derive trigonometric identities such as

$$\cos(\theta + \vartheta) = \cos\theta\cos\vartheta - \sin\theta\sin\vartheta,$$

and then show that

$$\begin{aligned} 2\sin\theta\sin\varphi &= \cos(\theta-\varphi) - \cos(\theta+\varphi), \\ 2\sin\theta\cos\varphi &= \sin(\theta+\varphi) + \sin(\theta-\varphi). \end{aligned}$$

This calculation connects the solution given by d'Alembert in terms of traveling waves and the solution in terms of superposition of standing waves.

[5] A sequence of functions $\{f_n(z)\}_{n=1}^{\infty}$ is said to be uniformly convergent on a set S if there exists a function f on S so that for every $\epsilon > 0$ there is an integer N such that $|f_n(z) - f(z)| < \epsilon$ whenever $n > N$ and $z \in S$.

5. Verify that $f(x) = e^{inx}$ is periodic with period 2π and that

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{inx}\,dx = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0. \end{cases}$$

Use this fact to prove that if $n, m \geq 1$ we have

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\,\cos mx\,dx = \begin{cases} 0 & \text{if } n \neq m, \\ 1 & n = m, \end{cases}$$

and similarly

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \sin nx\,\sin mx\,dx = \begin{cases} 0 & \text{if } n \neq m, \\ 1 & n = m. \end{cases}$$

Finally, show that

$$\int_{-\pi}^{\pi} \sin nx\,\cos mx\,dx = 0 \qquad \text{for any } n, m.$$

[Hint: Calculate $e^{inx}e^{-imx} + e^{inx}e^{imx}$ and $e^{inx}e^{-imx} - e^{inx}e^{imx}$.]

6. Prove that if f is a twice continuously differentiable function on $\mathbb{R}$ which is a solution of the equation

$$f''(t) + c^2 f(t) = 0,$$

then there exist constants a and b such that

$$f(t) = a \cos ct + b \sin ct.$$

This can be done by differentiating the two functions $g(t) = f(t) \cos ct - c^{-1} f'(t) \sin ct$ and $h(t) = f(t) \sin ct + c^{-1} f'(t) \cos ct$.

7. Show that if a and b are real, then one can write

$$a\cos ct + b\sin ct = A\cos(ct - \varphi),$$

where $A = \sqrt{a^2 + b^2}$, and φ is chosen so that

$$\cos\varphi = \frac{a}{\sqrt{a^2 + b^2}} \quad \text{and} \quad \sin\varphi = \frac{b}{\sqrt{a^2 + b^2}}.$$

8. Suppose F is a function on (a, b) with two continuous derivatives. Show that whenever x and $x + h$ belong to (a, b), one may write

$$F(x+h) = F(x) + hF'(x) + \frac{h^2}{2}F''(x) + h^2\varphi(h)\,,$$

where $\varphi(h) \to 0$ as $h \to 0$.

Deduce that

$$\frac{F(x+h) + F(x-h) - 2F(x)}{h^2} \to F''(x) \qquad \text{as } h \to 0.$$

[Hint: This is simply a Taylor expansion. It may be obtained by noting that

$$F(x+h) - F(x) = \int_x^{x+h} F'(y)\,dy,$$

and then writing $F'(y) = F'(x) + (y-x)F''(x) + (y-x)\psi(y-x)$, where $\psi(h) \to 0$ as $h \to 0$.]

9. In the case of the plucked string, use the formula for the Fourier sine coefficients to show that

$$A_m = \frac{2h}{m^2} \frac{\sin mp}{p(\pi - p)}.$$

For what position of p are the second, fourth, ... harmonics missing? For what position of p are the third, sixth, ... harmonics missing?

10. Show that the expression of the Laplacian

$$\triangle = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

is given in polar coordinates by the formula

$$\triangle = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}.$$

Also, prove that

$$\left|\frac{\partial u}{\partial x}\right|^2 + \left|\frac{\partial u}{\partial y}\right|^2 = \left|\frac{\partial u}{\partial r}\right|^2 + \frac{1}{r^2}\left|\frac{\partial u}{\partial \theta}\right|^2.$$

11. Show that if $n \in \mathbb{Z}$ the only solutions of the differential equation

$$r^2F''(r) + rF'(r) - n^2F(r) = 0,$$

which are twice differentiable when $r > 0$, are given by linear combinations of r^n and r^{-n} when $n \neq 0$, and 1 and $\log r$ when $n = 0$.

[Hint: If F solves the equation, write $F(r) = g(r)r^n$, find the equation satisfied by g, and conclude that $rg'(r) + 2ng(r) = c$ where c is a constant.]

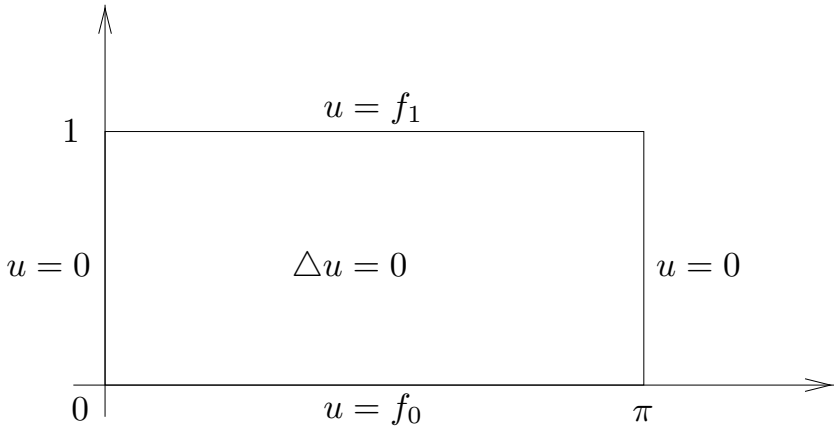


Figure 11. Dirichlet problem in a rectangle

Chapter 2. Basic Properties of Fourier Series

1. Suppose f is 2π-periodic and integrable on any finite interval. Prove that if $a, b \in \mathbb{R}$, then

$$\int_a^b f(x)\,dx = \int_{a+2\pi}^{b+2\pi} f(x)\,dx = \int_{a-2\pi}^{b-2\pi} f(x)\,dx.$$

Also prove that

$$\int_{-\pi}^{\pi} f(x+a)\,dx = \int_{-\pi}^{\pi} f(x)\,dx = \int_{-\pi+a}^{\pi+a} f(x)\,dx.$$

2. In this exercise we show how the symmetries of a function imply certain properties of its Fourier coefficients. Let f be a 2π-periodic Riemann integrable function defined on $\mathbb{R}$.

(a) Show that the Fourier series of the function f can be written as

$$f(\theta) \sim \hat{f}(0) + \sum_{n\geq 1} [\hat{f}(n) + \hat{f}(-n)] \cos n\theta + i[\hat{f}(n) - \hat{f}(-n)] \sin n\theta.$$

(b) Prove that if f is even, then $\hat{f}(n) = \hat{f}(-n)$, and we get a cosine series.

(c) Prove that if f is odd, then $\hat{f}(n) = -\hat{f}(-n)$, and we get a sine series.

(d) Suppose that $f(\theta + \pi) = f(\theta)$ for all $\theta \in \mathbb{R}$. Show that $\hat{f}(n) = 0$ for all odd n.

(e) Show that f is real-valued if and only if $\overline{\hat{f}(n)} = \hat{f}(-n)$ for all n.

3. We return to the problem of the plucked string discussed in Chapter 1. Show that the initial condition f is *equal* to its Fourier sine series

$$f(x) = \sum_{m=1}^{\infty} A_m \sin mx \qquad \text{with} \qquad A_m = \frac{2h}{m^2} \frac{\sin mp}{p(\pi - p)}.$$

[Hint: Note that $|A_m| \leq C/m^2$.]

4. Consider the 2π-periodic odd function defined on $[0, \pi]$ by $f(\theta) = \theta(\pi - \theta)$.

(a) Draw the graph of f.

(b) Compute the Fourier coefficients of f, and show that

$$f(\theta) = \frac{8}{\pi} \sum_{k \text{ odd} \geq 1} \frac{\sin k\theta}{k^3}.$$

5. On the interval $[-\pi, \pi]$ consider the function

$$f(\theta) = \begin{cases} 0 & \text{if } |\theta| > \delta, \\ 1 - |\theta|/\delta & \text{if } |\theta| \leq \delta. \end{cases}$$

Thus the graph of f has the shape of a triangular tent. Show that

$$f(\theta) = \frac{\delta}{2\pi} + 2\sum_{n=1}^{\infty} \frac{1 - \cos n\delta}{n^2 \pi \delta} \cos n\theta.$$

6. Let f be the function defined on $[-\pi, \pi]$ by $f(\theta) = |\theta|$.

(a) Draw the graph of f.

(b) Calculate the Fourier coefficients of f, and show that

$$\hat{f}(n) = \begin{cases} \dfrac{\pi}{2} & \text{if } n = 0, \\ \dfrac{-1 + (-1)^n}{\pi n^2} & \text{if } n \neq 0. \end{cases}$$

(c) What is the Fourier series of f in terms of sines and cosines?

(d) Taking $\theta = 0$, prove that

$$\sum_{n \text{ odd } \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8} \qquad \text{and} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

See also Example 2 in Section 1.1.

7. Suppose $\{a_n\}_{n=1}^N$ and $\{b_n\}_{n=1}^N$ are two finite sequences of complex numbers. Let $B_k = \sum_{n=1}^k b_n$ denote the partial sums of the series $\sum b_n$ with the convention $B_0 = 0$.

(a) Prove the **summation by parts** formula

$$\sum_{n=M}^{N} a_n b_n = a_N B_N - a_M B_{M-1} - \sum_{n=M}^{N-1} (a_{n+1} - a_n) B_n.$$

(b) Deduce from this formula Dirichlet's test for convergence of a series: if the partial sums of the series $\sum b_n$ are bounded, and $\{a_n\}$ is a sequence of real numbers that decreases monotonically to 0, then $\sum a_n b_n$ converges.

8. Verify that $\frac{1}{2i}\sum_{n\neq 0}\frac{e^{inx}}{n}$ is the Fourier series of the 2π-periodic **sawtooth** function illustrated in Figure 6, defined by $f(0)=0$, and

$$f(x)=\begin{cases} -\frac{\pi}{2}-\frac{x}{2} & \text{if } -\pi<x<0,\\ \\ \frac{\pi}{2}-\frac{x}{2} & \text{if } 0<x<\pi. \end{cases}$$

Note that this function is not continuous. Show that nevertheless, the series converges for every x (by which we mean, as usual, that the symmetric partial sums of the series converge). In particular, the value of the series at the origin, namely 0, is the average of the values of $f(x)$ as x approaches the origin from the left and the right.

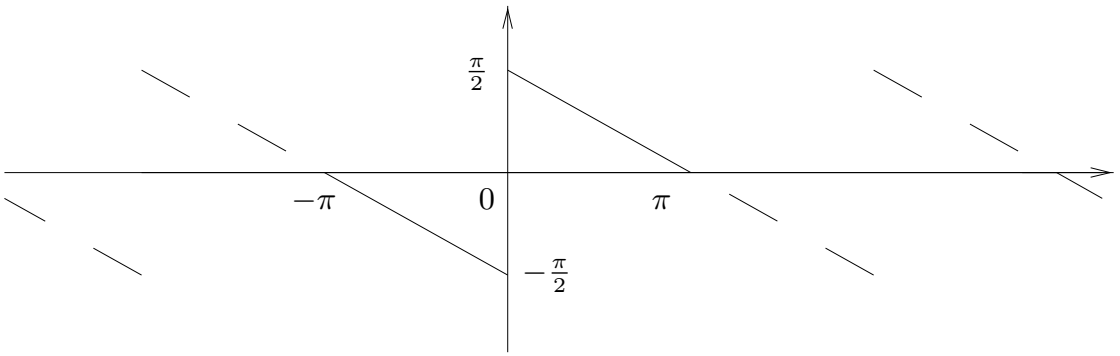


Figure 6. The sawtooth function

[Hint: Use Dirichlet's test for convergence of a series $\sum a_n b_n$.]

9. Let $f(x) = \chi_{[a,b]}(x)$ be the characteristic function of the interval $[a, b] \subset [-\pi, \pi]$, that is,

$$\chi_{[a,b]}(x) = \begin{cases} 1 & \text{if } x \in [a, b], \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that the Fourier series of f is given by

$$f(x) \sim \frac{b-a}{2\pi} + \sum_{n \neq 0} \frac{e^{-ina} - e^{-inb}}{2\pi i n} \, e^{inx}.$$

The sum extends over all positive and negative integers excluding 0.

(b) Show that if $a \neq -\pi$ or $b \neq \pi$ and $a \neq b$, then the Fourier series does not converge absolutely for any x. [Hint: It suffices to prove that for many values of n one has $|\sin n\theta_0| \geq c > 0$ where $\theta_0 = (b-a)/2$.]

(c) However, prove that the Fourier series converges at every point x. What happens if $a = -\pi$ and $b = \pi$?

10. Suppose f is a periodic function of period 2π which belongs to the class C^k. Show that

$$\hat{f}(n) = O(1/|n|^k) \qquad \text{as } |n| \to \infty.$$

This notation means that there exists a constant C such $|\hat{f}(n)| \leq C/|n|^k$. We could also write this as $|n|^k \hat{f}(n) = O(1)$, where $O(1)$ means bounded.

[Hint: Integrate by parts.]

11. Suppose that $\{f_k\}_{k=1}^{\infty}$ is a sequence of Riemann integrable functions on the interval $[0,1]$ such that

$$\int_0^1 |f_k(x) - f(x)|\, dx \to 0 \qquad \text{as } k \to \infty.$$

Show that $\hat{f}_k(n) \to \hat{f}(n)$ uniformly in n as $k \to \infty$.

12. Prove that if a series of complex numbers $\sum c_n$ converges to s, then $\sum c_n$ is Cesàro summable to s.

[Hint: Assume $s_n \to 0$ as $n \to \infty$.]

13. The purpose of this exercise is to prove that Abel summability is stronger than the standard or Cesàro methods of summation.

(a) Show that if the series $\sum_{n=1}^{\infty} c_n$ of complex numbers converges to a finite limit s, then the series is Abel summable to s. [Hint: Why is it enough to prove the theorem when $s = 0$? Assuming $s = 0$, show that if $s_N = c_1 + \cdots + c_N$, then $\sum_{n=1}^{N} c_n r^n = (1-r)\sum_{n=1}^{N} s_n r^n + s_N r^{N+1}$. Let $N \to \infty$ to show that

$$\sum c_n r^n = (1-r)\sum s_n r^n.$$

Finally, prove that the right-hand side converges to 0 as $r \to 1$.]

(b) However, show that there exist series which are Abel summable, but that do not converge. [Hint: Try $c_n = (-1)^n$. What is the Abel limit of $\sum c_n$?]

(c) Argue similarly to prove that if a series $\sum_{n=1}^{\infty} c_n$ is Cesàro summable to σ, then it is Abel summable to σ. [Hint: Note that

$$\sum_{n=1}^{\infty} c_n r^n = (1-r)^2 \sum_{n=1}^{\infty} n\sigma_n r^n,$$

and assume $\sigma = 0$.]

(d) Give an example of a series that is Abel summable but not Cesàro summable. [Hint: Try $c_n = (-1)^{n-1} n$. Note that if $\sum c_n$ is Cesàro summable, then c_n/n tends to 0.]

The results above can be summarized by the following implications about series:

$$\text{convergent} \Longrightarrow \text{Cesàro summable} \Longrightarrow \text{Abel summable},$$

and the fact that none of the arrows can be reversed.

14. This exercise deals with a theorem of Tauber which says that under an additional condition on the coefficients c_n, the above arrows can be reversed.

(a) If $\sum c_n$ is Cesàro summable to σ and $c_n = o(1/n)$ (that is, $nc_n \to 0$), then $\sum c_n$ converges to σ. [Hint: $s_n - \sigma_n = [(n-1)c_n + \cdots + c_2]/n$.]

(b) The above statement holds if we replace Cesàro summable by Abel summable. [Hint: Estimate the difference between $\sum_{n=1}^{N} c_n$ and $\sum_{n=1}^{N} c_n r^n$ where $r = 1 - 1/N$.]

15. Prove that the Fejér kernel is given by

$$F_N(x) = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}.$$

[Hint: Remember that $NF_N(x) = D_0(x) + \cdots + D_{N-1}(x)$ where $D_n(x)$ is the Dirichlet kernel. Therefore, if $\omega = e^{ix}$ we have

$$NF_N(x) = \sum_{n=0}^{N-1} \frac{\omega^{-n} - \omega^{n+1}}{1 - \omega}.]$$

16. The Weierstrass approximation theorem states: Let f be a continuous function on the closed and bounded interval $[a, b] \subset \mathbb{R}$. Then, for any $\epsilon > 0$, there exists a polynomial P such that

$$\sup_{x \in [a,b]} |f(x) - P(x)| < \epsilon.$$

Prove this by applying Corollary 5.4 of Fejér's theorem and using the fact that the exponential function e^{ix} can be approximated by polynomials uniformly on any interval.

17. In Section 5.4 we proved that the Abel means of f converge to f at all points of continuity, that is,

$$\lim_{r\to 1} A_r(f)(\theta) = \lim_{r\to 1}(P_r * f)(\theta) = f(\theta), \qquad \text{with } 0 < r < 1,$$

whenever f is continuous at θ. In this exercise, we will study the behavior of $A_r(f)(\theta)$ at certain points of discontinuity.

An integrable function is said to have a **jump discontinuity** at θ if the two limits

$$\lim_{\substack{h\,\to\,0\\ h\,>\,0}} f(\theta+h) = f(\theta^+) \qquad \text{and} \qquad \lim_{\substack{h\,\to\,0\\ h\,>\,0}} f(\theta-h) = f(\theta^-)$$

exist.

(a) Prove that if f has a jump discontinuity at θ, then

$$\lim_{r\to 1} A_r(f)(\theta) = \frac{f(\theta^+) + f(\theta^-)}{2}, \qquad \text{with } 0 \le r < 1.$$

[Hint: Explain why $\frac{1}{2\pi}\int_{-\pi}^{0} P_r(\theta)\,d\theta = \frac{1}{2\pi}\int_{0}^{\pi} P_r(\theta)\,d\theta = \frac{1}{2}$, then modify the proof given in the text.]

(b) Using a similar argument, show that if f has a jump discontinuity at θ, the Fourier series of f at θ is Cesàro summable to $\frac{f(\theta^+)+f(\theta^-)}{2}$.

18. If $P_r(\theta)$ denotes the Poisson kernel, show that the function

$$u(r,\theta) = \frac{\partial P_r}{\partial \theta},$$

defined for $0 \leq r < 1$ and $\theta \in \mathbb{R}$, satisfies:

(i) $\triangle u = 0$ in the disc.

(ii) $\lim_{r\to 1} u(r,\theta) = 0$ for each θ.

However, u is not identically zero.

19. Solve Laplace's equation $\triangle u = 0$ in the semi infinite strip

$$S = \{(x, y) : 0 < x < 1,\ 0 < y\},$$

subject to the following boundary conditions

$$\begin{cases} u(0, y) = 0 & \text{when } 0 \leq y, \\ u(1, y) = 0 & \text{when } 0 \leq y, \\ u(x, 0) = f(x) & \text{when } 0 \leq x \leq 1 \end{cases}$$

where f is a given function, with of course $f(0) = f(1) = 0$. Write

$$f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x)$$

and expand the general solution in terms of the special solutions given by

$$u_n(x, y) = e^{-n\pi y} \sin(n\pi x).$$

Express u as an integral involving f, analogous to the Poisson integral formula (6).

20. Consider the Dirichlet problem in the annulus defined by $\{(r,\theta) : \rho < r < 1\}$, where $0 < \rho < 1$ is the inner radius. The problem is to solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$$

subject to the boundary conditions

$$\begin{cases} u(1,\theta) = f(\theta), \\ u(\rho,\theta) = g(\theta), \end{cases}$$

where f and g are given continuous functions.

Arguing as we have previously for the Dirichlet problem in the disc, we can hope to write

$$u(r,\theta) = \sum c_n(r) e^{in\theta}$$

with $c_n(r) = A_n r^n + B_n r^{-n}$, $n \neq 0$. Set

$$f(\theta) \sim \sum a_n e^{in\theta} \qquad \text{and} \qquad g(\theta) \sim \sum b_n e^{in\theta}.$$

We want $c_n(1) = a_n$ and $c_n(\rho) = b_n$. This leads to the solution

$$\begin{aligned} u(r,\theta) = &\sum_{n\neq 0} \left(\frac{1}{\rho^n - \rho^{-n}}\right) \left[((\rho/r)^n - (r/\rho)^n)\, a_n + (r^n - r^{-n}) b_n\right] e^{in\theta} \\ &+ a_0 + (b_0 - a_0)\frac{\log r}{\log \rho}. \end{aligned}$$

Show that as a result we have

$$u(r,\theta) - (P_r * f)(\theta) \to 0 \qquad \text{as } r \to 1 \text{ uniformly in } \theta,$$

and

$$u(r,\theta) - (P_{\rho/r} * g)(\theta) \to 0 \qquad \text{as } r \to \rho \text{ uniformly in } \theta.$$

Chapter 3. Convergence of Fourier Series

1. Show that the first two examples of inner product spaces, namely $\mathbb{R}^d$ and $\mathbb{C}^d$, are complete.

[Hint: Every Cauchy sequence in $\mathbb{R}$ has a limit.]

2. Prove that the vector space $\ell^2(\mathbb{Z})$ is complete.

[Hint: Suppose $A_k = \{a_{k,n}\}_{n\in\mathbb{Z}}$ with $k = 1, 2, \ldots$ is a Cauchy sequence. Show that for each n, $\{a_{k,n}\}_{k=1}^{\infty}$ is a Cauchy sequence of complex numbers, therefore it converges to a limit, say b_n. By taking partial sums of $\|A_k - A_{k'}\|$ and letting $k' \to \infty$, show that $\|A_k - B\| \to 0$ as $k \to \infty$, where $B = (\ldots, b_{-1}, b_0, b_1, \ldots)$. Finally, prove that $B \in \ell^2(\mathbb{Z})$.]

3. Construct a sequence of integrable functions $\{f_k\}$ on $[0, 2\pi]$ such that

$$\lim_{k\to\infty} \frac{1}{2\pi} \int_0^{2\pi} |f_k(\theta)|^2 \, d\theta = 0$$

but $\lim_{k\to\infty} f_k(\theta)$ fails to exist for any θ.

[Hint: Choose a sequence of intervals $I_k \subset [0, 2\pi]$ whose lengths tend to 0, and so that each point belongs to infinitely many of them; then let $f_k = \chi_{I_k}$.]

4. Recall the vector space $\mathcal{R}$ of integrable functions, with its inner product and norm

$$\|f\| = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(x)|^2 \, dx \right)^{1/2}.$$

(a) Show that there exist non-zero integrable functions f for which $\|f\| = 0$.

(b) However, show that if $f \in \mathcal{R}$ with $\|f\| = 0$, then $f(x) = 0$ whenever f is continuous at x.

(c) Conversely, show that if $f \in \mathcal{R}$ vanishes at all of its points of continuity, then $\|f\| = 0$.

5. Let

$$f(\theta) = \begin{cases} 0 & \text{for } \theta = 0 \\ \log(1/\theta) & \text{for } 0 < \theta \le 2\pi, \end{cases}$$

and define a sequence of functions in $\mathcal{R}$ by

$$f_n(\theta) = \begin{cases} 0 & \text{for } 0 \le \theta \le 1/n \\ f(\theta) & \text{for } 1/n < \theta \le 2\pi. \end{cases}$$

Prove that $\{f_n\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathcal{R}$. However, f does not belong to $\mathcal{R}$.

[Hint: Show that $\int_a^b (\log\theta)^2\, d\theta \to 0$ if $0 < a < b$ and $b \to 0$, by using the fact that the derivative of $\theta(\log\theta)^2 - 2\theta\log\theta + 2\theta$ is equal to $(\log\theta)^2$.]

6. Consider the sequence $\{a_k\}_{k=-\infty}^{\infty}$ defined by

$$a_k = \begin{cases} 1/k & \text{if } k \geq 1 \\ 0 & \text{if } k \leq 0. \end{cases}$$

Note that $\{a_k\} \in \ell^2(\mathbb{Z})$, but that no Riemann integrable function has k^{th} Fourier coefficient equal to a_k for all k.

7. Show that the trigonometric series

$$\sum_{n\geq 2} \frac{1}{\log n} \sin nx$$

converges for every x, yet it is not the Fourier series of a Riemann integrable function.

The same is true for $\sum \frac{\sin nx}{n^\alpha}$ for $0 < \alpha < 1$, but the case $1/2 < \alpha < 1$ is more difficult. See Problem 1.

8. Exercise 6 in Chapter 2 dealt with the sums

$$\sum_{n \text{ odd } \geq 1} \frac{1}{n^2} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Similar sums can be derived using the methods of this chapter.

(a) Let f be the function defined on $[-\pi, \pi]$ by $f(\theta) = |\theta|$. Use Parseval's identity to find the sums of the following two series:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^4} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^4}.$$

In fact, they are $\pi^4/96$ and $\pi^4/90$, respectively.

(b) Consider the 2π-periodic odd function defined on $[0, \pi]$ by $f(\theta) = \theta(\pi - \theta)$. Show that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^6} = \frac{\pi^6}{960} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}.$$

Remark. The general expression when k is even for $\sum_{n=1}^{\infty} 1/n^k$ in terms of π^k is given in Problem 4. However, finding a formula for the sum $\sum_{n=1}^{\infty} 1/n^3$, or more generally $\sum_{n=1}^{\infty} 1/n^k$ with k odd, is a famous unresolved question.

9. Show that for α not an integer, the Fourier series of

$$\frac{\pi}{\sin \pi\alpha} e^{i(\pi - x)\alpha}$$

on $[0, 2\pi]$ is given by

$$\sum_{n=-\infty}^{\infty} \frac{e^{inx}}{n+\alpha}.$$

Apply Parseval's formula to show that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{(\sin \pi\alpha)^2}.$$

10. Consider the example of a vibrating string which we analyzed in Chapter 1. The displacement $u(x,t)$ of the string at time t satisfies the wave equation

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c^2 = \tau/\rho.$$

The string is subject to the initial conditions

$$u(x,0) = f(x) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(x,0) = g(x),$$

where we assume that $f \in C^1$ and g is continuous. We define the total **energy** of the string by

$$E(t) = \frac{1}{2}\rho \int_0^L \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2}\tau \int_0^L \left(\frac{\partial u}{\partial x}\right)^2 dx.$$

The first term corresponds to the "kinetic energy" of the string (in analogy with $(1/2)mv^2$, the kinetic energy of a particle of mass m and velocity v), and the second term corresponds to its "potential energy."

Show that the total energy of the string is conserved, in the sense that $E(t)$ is constant. Therefore,

$$E(t) = E(0) = \frac{1}{2}\rho \int_0^L g(x)^2\,dx + \frac{1}{2}\tau \int_0^L f'(x)^2\,dx.$$

11. The inequalities of Wirtinger and Poincaré establish a relationship between the norm of a function and that of its derivative.

(a) If f is T-periodic, continuous, and piecewise C^1 with $\int_0^T f(t)\,dt = 0$, show that

$$\int_0^T |f(t)|^2\,dt \le \frac{T^2}{4\pi^2}\int_0^T |f'(t)|^2\,dt,$$

with equality if and only if $f(t) = A\sin(2\pi t/T) + B\cos(2\pi t/T)$. [Hint: Apply Parseval's identity.]

(b) If f is as above and g is just C^1 and T-periodic, prove that

$$\left|\int_0^T \overline{f(t)}g(t)\,dt\right|^2 \le \frac{T^2}{4\pi^2}\int_0^T |f(t)|^2\,dt\int_0^T |g'(t)|^2\,dt.$$

(c) For any compact interval $[a,b]$ and any continuously differentiable function f with $f(a) = f(b) = 0$, show that

$$\int_a^b |f(t)|^2\,dt \le \frac{(b-a)^2}{\pi^2}\int_a^b |f'(t)|^2\,dt.$$

Discuss the case of equality, and prove that the constant $(b-a)^2/\pi^2$ cannot be improved. [Hint: Extend f to be odd with respect to a and periodic of period $T = 2(b-a)$ so that its integral over an interval of length T is 0. Apply part a) to get the inequality, and conclude that equality holds if and only if $f(t) = A\sin(\pi\frac{t-a}{b-a})$].

12. Prove that $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$.

[Hint: Start with the fact that the integral of $D_N(\theta)$ equals 2π, and note that the difference $(1/\sin(\theta/2)) - 2/\theta$ is continuous on $[-\pi, \pi]$. Apply the Riemann-Lebesgue lemma.]

13. Suppose that f is periodic and of class C^k. Show that

$$\hat{f}(n) = o(1/|n|^k),$$

that is, $|n|^k \hat{f}(n)$ goes to 0 as $|n| \to \infty$. This is an improvement over Exercise 10 in Chapter 2.

[Hint: Use the Riemann-Lebesgue lemma.]

14. Prove that the Fourier series of a continuously differentiable function f on the circle is absolutely convergent.

[Hint: Use the Cauchy-Schwarz inequality and Parseval's identity for f'.]

15. Let f be 2π-periodic and Riemann integrable on $[-\pi, \pi]$.

(a) Show that

$$\hat{f}(n) = -\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x+\pi/n)e^{-inx}\,dx$$

hence

$$\hat{f}(n) = \frac{1}{4\pi}\int_{-\pi}^{\pi} [f(x) - f(x+\pi/n)]e^{-inx}\,dx.$$

(b) Now assume that f satisfies a Hölder condition of order α, namely

$$|f(x+h) - f(x)| \leq C|h|^{\alpha}$$

for some $0 < \alpha \leq 1$, some $C > 0$, and all x, h. Use part a) to show that

$$\hat{f}(n) = O(1/|n|^{\alpha}).$$

(c) Prove that the above result cannot be improved by showing that the function

$$f(x) = \sum_{k=0}^{\infty} 2^{-k\alpha} e^{i2^k x},$$

where $0 < \alpha < 1$, satisfies

$$|f(x+h) - f(x)| \leq C|h|^{\alpha},$$

and $\hat{f}(N) = 1/N^{\alpha}$ whenever $N = 2^k$.

[Hint: For (c), break up the sum as follows $f(x+h) - f(x) = \sum_{2^k \leq 1/|h|} + \sum_{2^k > 1/|h|}$. To estimate the first sum use the fact that $|1 - e^{i\theta}| \leq |\theta|$ whenever θ is small. To estimate the second sum, use the obvious inequality $|e^{ix} - e^{iy}| \leq 2$.]

16. Let f be a 2π-periodic function which satisfies a Lipschitz condition with constant K; that is,

$$|f(x) - f(y)| \leq K|x - y| \qquad \text{for all } x, y.$$

This is simply the Hölder condition with $\alpha = 1$, so by the previous exercise, we see that $\hat{f}(n) = O(1/|n|)$. Since the harmonic series $\sum 1/n$ diverges, we cannot say anything (yet) about the absolute convergence of the Fourier series of f. The outline below actually proves that the Fourier series of f converges absolutely and uniformly.

(a) For every positive h we define $g_h(x) = f(x + h) - f(x - h)$. Prove that

$$\frac{1}{2\pi} \int_0^{2\pi} |g_h(x)|^2 \, dx = \sum_{n=-\infty}^{\infty} 4|\sin nh|^2 |\hat{f}(n)|^2,$$

and show that

$$\sum_{n=-\infty}^{\infty} |\sin nh|^2 |\hat{f}(n)|^2 \leq K^2 h^2.$$

(b) Let p be a positive integer. By choosing $h = \pi/2^{p+1}$, show that

$$\sum_{2^{p-1} < |n| \leq 2^p} |\hat{f}(n)|^2 \leq \frac{K^2 \pi^2}{2^{2p+1}}.$$

(c) Estimate $\sum_{2^{p-1} < |n| \leq 2^p} |\hat{f}(n)|$, and conclude that the Fourier series of f converges absolutely, hence uniformly. [Hint: Use the Cauchy-Schwarz inequality to estimate the sum.]

(d) In fact, modify the argument slightly to prove Bernstein's theorem: If f satisfies a Hölder condition of order $\alpha > 1/2$, then the Fourier series of f converges absolutely.

17. If f is a bounded monotonic function on $[-\pi, \pi]$, then

$$\hat{f}(n) = O(1/|n|).$$

[Hint: One may assume that f is increasing, and say $|f| \leq M$. First check that the Fourier coefficients of the characteristic function of $[a, b]$ satisfy $O(1/|n|)$. Now show that a sum of the form

$$\sum_{k=1}^{N} \alpha_k \chi_{[a_k, a_{k+1}]}(x)$$

with $-\pi = a_1 < a_2 < \cdots < a_N < a_{N+1} = \pi$ and $-M \leq \alpha_1 \leq \cdots \leq \alpha_N \leq M$ has Fourier coefficients that are $O(1/|n|)$ uniformly in N. Summing by parts one gets a telescopic sum $\sum(\alpha_{k+1} - \alpha_k)$ which can be bounded by $2M$. Now approximate f by functions of the above type.]

18. Here are a few things we have learned about the decay of Fourier coefficients:

(a) if f is of class C^k, then $\hat{f}(n) = o(1/|n|^k)$;

(b) if f is Lipschitz, then $\hat{f}(n) = O(1/|n|)$;

(c) if f is monotonic, then $\hat{f}(n) = O(1/|n|)$;

(d) if f is satisfies a Hölder condition with exponent α where $0 < \alpha < 1$, then $\hat{f}(n) = O(1/|n|^\alpha)$;

(e) if f is merely Riemann integrable, then $\sum |\hat{f}(n)|^2 < \infty$ and therefore $\hat{f}(n) = o(1)$.

Nevertheless, show that the Fourier coefficients of a continuous function can tend to 0 arbitrarily slowly by proving that for every sequence of nonnegative real numbers $\{\epsilon_n\}$ converging to 0, there exists a continuous function f such that $|\hat{f}(n)| \geq \epsilon_n$ for infinitely many values of n.

[Hint: Choose a subsequence $\{\epsilon_{n_k}\}$ so that $\sum_k \epsilon_{n_k} < \infty$.]

19. Give another proof that the sum $\sum_{0<|n|\le N} e^{inx}/n$ is uniformly bounded in N and $x \in [-\pi, \pi]$ by using the fact that

$$\frac{1}{2i} \sum_{0<|n|\le N} \frac{e^{inx}}{n} = \sum_{n=1}^{N} \frac{\sin nx}{n} = \frac{1}{2} \int_0^x (D_N(t) - 1)\, dt,$$

where D_N is the Dirichlet kernel. Now use the fact that $\int_0^\infty \frac{\sin t}{t}\, dt < \infty$ which was proved in Exercise 12.

20. Let $f(x)$ denote the sawtooth function defined by $f(x) = (\pi - x)/2$ on the interval $(0, 2\pi)$ with $f(0) = 0$ and extended by periodicity to all of $\mathbb{R}$. The Fourier series of f is

$$f(x) \sim \frac{1}{2i} \sum_{|n|\neq 0} \frac{e^{inx}}{n} = \sum_{n=1}^{\infty} \frac{\sin nx}{n},$$

and f has a jump discontinuity at the origin with

$$f(0^+) = \frac{\pi}{2}, \qquad f(0^-) = -\frac{\pi}{2}, \qquad \text{and hence} \qquad f(0^+) - f(0^-) = \pi.$$

Show that

$$\max_{0<x\leq \pi/N} S_N(f)(x) - \frac{\pi}{2} = \int_0^{\pi} \frac{\sin t}{t}\, dt - \frac{\pi}{2},$$

which is roughly 9% of the jump π. This result is a manifestation of Gibbs's phenomenon which states that near a jump discontinuity, the Fourier series of a function overshoots (or undershoots) it by approximately 9% of the jump.

[Hint: Use the expression for $S_N(f)$ given in Exercise 19.]

Chapter 4. Some Applications of Fourier Series

1. Let $\gamma : [a, b] \to \mathbb{R}^2$ be a parametrization for the closed curve Γ.

(a) Prove that γ is a parametrization by arc-length if and only if the length of the curve from $\gamma(a)$ to $\gamma(s)$ is precisely $s - a$, that is,

$$\int_a^s |\gamma'(t)|\, dt = s - a.$$

(b) Prove that any curve Γ admits a parametrization by arc-length. [Hint: If η is any parametrization, let $h(s) = \int_a^s |\eta'(t)|\, dt$ and consider $\gamma = \eta \circ h^{-1}$.]

2. Suppose $\gamma : [a, b] \to \mathbb{R}^2$ is a parametrization for a closed curve Γ, with $\gamma(t) = (x(t), y(t))$.

(a) Show that

$$\frac{1}{2}\int_a^b (x(s)y'(s) - y(s)x'(s))\,ds = \int_a^b x(s)y'(s)\,ds = -\int_a^b y(s)x'(s)\,ds.$$

(b) Define the **reverse parametrization** of γ by $\gamma^- : [a, b] \to \mathbb{R}^2$ with $\gamma^-(t) = \gamma(b + a - t)$. The image of γ^- is precisely Γ, except that the points $\gamma^-(t)$ and $\gamma(t)$ travel in opposite directions. Thus γ^- "reverses" the orientation of the curve. Prove that

$$\int_\gamma (x\,dy - y\,dx) = -\int_{\gamma^-} (x\,dy - y\,dx).$$

In particular, we may assume (after a possible change in orientation) that

$$\mathcal{A} = \frac{1}{2}\int_a^b (x(s)y'(s) - y(s)x'(s))\,ds = \int_a^b x(s)y'(s)\,ds.$$

3. Suppose Γ is a curve in the plane, and that there exists a set of coordinates x and y so that the x-axis divides the curve into the union of the graph of two continuous functions $y = f(x)$ and $y = g(x)$ for $0 \leq x \leq 1$, and with $f(x) \geq g(x)$ (see Figure 6). Let Ω denote the region between the graphs of these two functions:

$$\Omega = \{(x, y) : \ 0 \leq x \leq 1 \ \text{ and } \ g(x) \leq y \leq f(x)\}.$$

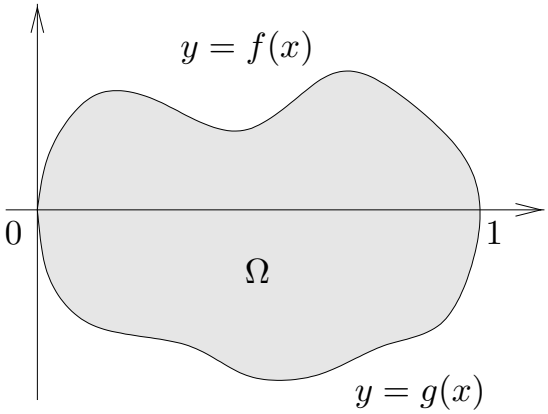


Figure 6. Simple version of the area formula

With the familiar interpretation that the integral $\int h(x)\,dx$ gives the area under the graph of the function h, we see that the area of Ω is $\int_0^1 f(x)\,dx - \int_0^1 g(x)\,dx$. Show that this definition coincides with the area formula $\mathcal{A}$ given in the text, that is,

$$\int_0^1 f(x)\,dx - \int_0^1 g(x)\,dx = \left| -\int_\Gamma y\,dx \right| = \mathcal{A}.$$

Also, note that if the orientation of the curve is chosen so that Ω "lies to the left" of Γ, then the above formula holds without the absolute value signs.

This formula generalizes to any set that can be written as a finite union of domains like Ω above.

4. Observe that with the definition of ℓ and $\mathcal{A}$ given in the text, the isoperimetric inequality continues to hold (with the same proof) even when Γ is not simple.

Show that this stronger version of the isoperimetric inequality is equivalent to Wirtinger's inequality, which says that if f is 2π-periodic, of class C^1, and satisfies $\int_0^{2\pi} f(t)\,dt = 0$, then

$$\int_0^{2\pi} |f(t)|^2\,dt \leq \int_0^{2\pi} |f'(t)|^2\,dt$$

with equality if and only if $f(t) = A\sin t + B\cos t$ (Exercise 11, Chapter 3).

[Hint: In one direction, note that if the length of the curve is 2π and γ is an appropriate arc-length parametrization, then

$$2(\pi - \mathcal{A}) = \int_0^{2\pi} \left[x'(s) + y(s)\right]^2\,ds + \int_0^{2\pi} (y'(s)^2 - y(s)^2)\,ds.$$

A change of coordinates will guarantee $\int_0^{2\pi} y(s)\,ds = 0$. For the other direction, start with a real-valued f satisfying all the hypotheses of Wirtinger's inequality, and construct g, 2π-periodic and so that the term in brackets above vanishes.]

5. Prove that the sequence $\{\gamma_n\}_{n=1}^{\infty}$, where γ_n is the fractional part of

$$\left(\frac{1+\sqrt{5}}{2}\right)^n,$$

is not equidistributed in $[0,1]$.

[Hint: Show that $U_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$ is the solution of the difference equation $U_{r+1} = U_r + U_{r-1}$ with $U_0 = 2$ and $U_1 = 1$. The U_n satisfy the same difference equation as the Fibonacci numbers.]

6. Let $\theta = p/q$ be a rational number where p and q are relatively prime integers (that is, θ is in lowest form). We assume without loss of generality that $q > 0$. Define a sequence of numbers in $[0, 1)$ by $\xi_n = \langle n\theta \rangle$ where $\langle \cdot \rangle$ denotes the fractional part. Show that the sequence $\{\xi_1, \xi_2, \ldots\}$ is equidistributed on the points of the form

$$0,\ 1/q,\ 2/q,\ \ldots,\ (q-1)/q.$$

In fact, prove that for any $0 \leq a < q$, one has

$$\frac{\#\{n:\ 1 \leq n \leq N,\ \langle n\theta \rangle = a/q\}}{N} = \frac{1}{q} + O\left(\frac{1}{N}\right).$$

[Hint: For each integer $k \geq 0$, there exists a unique integer n with $kq \leq n < (k+1)q$ and so that $\langle n\theta \rangle = a/q$. Why can one assume $k = 0$? Prove the existence of n by using the fact[1] that if p and q are relatively prime, there exist integers x, y such that $xp + yq = 1$. Next, divide N by q with remainder, that is, write $N = \ell q + r$ where $0 \leq \ell$ and $0 \leq r < q$. Establish the inequalities

$$\ell \leq \#\{n:\ 1 \leq n \leq N,\ \langle n\theta \rangle = a/q\} \leq \ell + 1.]$$

[1]The elementary results in arithmetic used in this exercise can be found at the beginning of Chapter 8.

7. Prove the second part of Weyl's criterion: if a sequence of numbers $\xi_1, \xi_2, \ldots$ in $[0, 1)$ is equidistributed, then for all $k \in \mathbb{Z} - \{0\}$

$$\frac{1}{N} \sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0 \qquad \text{as } N \to \infty.$$

[Hint: It suffices to show that $\frac{1}{N} \sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f(x)\, dx$ for all continuous f. Prove this first when f is the characteristic function of an interval.]

8. Show that for any $a \neq 0$, and σ with $0 < \sigma < 1$, the sequence $\langle an^\sigma \rangle$ is equidistributed in $[0, 1)$.

[Hint: Prove that $\sum_{n=1}^{N} e^{2\pi i b n^\sigma} = O(N^\sigma) + O(N^{1-\sigma})$ if $b \neq 0$.] In fact, note the following

$$\sum_{n=1}^{N} e^{2\pi i b n^\sigma} - \int_1^N e^{2\pi i b x^\sigma}\, dx = O\left(\sum_{n=1}^{N} n^{-1+\sigma}\right).$$

9. In contrast with the result in Exercise 8, prove that $\langle a \log n \rangle$ is *not* equidistributed for any a.
[Hint: Compare the sum $\sum_{n=1}^{N} e^{2\pi i b \log n}$ with the corresponding integral.]

10. Suppose that f is a periodic function on $\mathbb{R}$ of period 1, and $\{\xi_n\}$ is a sequence which is equidistributed in $[0,1)$. Prove that:

(a) If f is continuous and satisfies $\int_0^1 f(x)\,dx = 0$, then

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(x+\xi_n) = 0 \qquad \text{uniformly in } x.$$

[Hint: Establish this result first for trigonometric polynomials.]

(b) If f is merely integrable on $[0,1]$ and satisfies $\int_0^1 f(x)\,dx = 0$, then

$$\lim_{N\to\infty} \int_0^1 \left| \frac{1}{N} \sum_{n=1}^{N} f(x+\xi_n) \right|^2 dx = 0.$$

11. Show that if $u(x,t) = (f * H_t)(x)$ where H_t is the heat kernel, and f is Riemann integrable, then

$$\int_0^1 |u(x,t) - f(x)|^2 \, dx \to 0 \qquad \text{as } t \to 0.$$

12. A change of variables in (8) leads to the solution

$$u(\theta,\tau) = \sum a_n e^{-n^2\tau} e^{in\theta} = (f * h_\tau)(\theta)$$

of the equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial \theta^2} \qquad \text{with } 0 \le \theta \le 2\pi \text{ and } \tau > 0,$$

with boundary condition $u(\theta,0) = f(\theta) \sim \sum a_n e^{in\theta}$. Here $h_\tau(\theta) = \sum_{n=-\infty}^{\infty} e^{-n^2\tau} e^{in\theta}$. This version of the heat kernel on $[0, 2\pi]$ is the analogue of the Poisson kernel, which can be written as $P_r(\theta) = \sum_{n=-\infty}^{\infty} e^{-|n|\tau} e^{in\theta}$ with $r = e^{-\tau}$ (and so $0 < r < 1$ corresponds to $\tau > 0$).

13. The fact that the kernel $H_t(x)$ is a good kernel, hence $u(x,t) \to f(x)$ at each point of continuity of f, is not easy to prove. This will be shown in the next chapter. However, one can prove directly that $H_t(x)$ is "peaked" at $x = 0$ as $t \to 0$ in the following sense:

(a) Show that $\int_{-1/2}^{1/2} |H_t(x)|^2\,dx$ is of the order of magnitude of $t^{-1/2}$ as $t \to 0$.

More precisely, prove that $t^{1/2} \int_{-1/2}^{1/2} |H_t(x)|^2\,dx$ converges to a non-zero limit as $t \to 0$.

(b) Prove that $\int_{-1/2}^{1/2} x^2 |H_t(x)|^2\,dx = O(t^{1/2})$ as $t \to 0$.

[Hint: For (a) compare the sum $\sum_{-\infty}^{\infty} e^{-cn^2 t}$ with the integral $\int_{-\infty}^{\infty} e^{-cx^2 t}\,dx$ where $c > 0$. For (b) use $x^2 \le C(\sin \pi x)^2$ for $-1/2 \le x \le 1/2$, and apply the mean value theorem to $e^{-cx^2 t}$.]

Chapter 5. The Fourier Transform on R

1. Corollary 2.3 in Chapter 2 leads to the following simplified version of the Fourier inversion formula. Suppose f is a continuous function supported on an interval $[-M, M]$, whose Fourier transform $\hat{f}$ is of moderate decrease.

(a) Fix L with $L/2 > M$, and show that $f(x) = \sum a_n(L) e^{2\pi i n x/L}$ where

$$a_n(L) = \frac{1}{L} \int_{-L/2}^{L/2} f(x) e^{-2\pi i n x/L}\, dx = \frac{1}{L} \hat{f}(n/L).$$

Alternatively, we may write $f(x) = \delta \sum_{n=-\infty}^{\infty} \hat{f}(n\delta) e^{2\pi i n \delta x}$ with $\delta = 1/L$.

(b) Prove that if F is continuous and of moderate decrease, then

$$\int_{-\infty}^{\infty} F(\xi)\, d\xi = \lim_{\substack{\delta \to 0 \\ \delta > 0}} \delta \sum_{n=-\infty}^{\infty} F(\delta n).$$

(c) Conclude that $f(x) = \displaystyle\int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi}\, d\xi$.

[Hint: For (a), note that the Fourier series of f on $[-L/2, L/2]$ converges absolutely. For (b), first approximate the integral by $\int_{-N}^{N} F$ and the sum by $\delta \sum_{|n| \le N/\delta} F(n\delta)$. Then approximate the second integral by Riemann sums.]

2. Let f and g be the functions defined by

$$f(x) = \chi_{[-1,1]}(x) = \begin{cases} 1 & \text{if } |x| \le 1, \\ 0 & \text{otherwise,} \end{cases} \qquad \text{and} \qquad g(x) = \begin{cases} 1 - |x| & \text{if } |x| \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Although f is not continuous, the integral defining its Fourier transform still makes sense. Show that

$$\hat{f}(\xi) = \frac{\sin 2\pi\xi}{\pi\xi} \qquad \text{and} \qquad \hat{g}(\xi) = \left(\frac{\sin \pi\xi}{\pi\xi}\right)^2,$$

with the understanding that $\hat{f}(0) = 2$ and $\hat{g}(0) = 1$.

3. The following exercise illustrates the principle that the decay of $\hat{f}$ is related to the continuity properties of f.

(a) Suppose that f is a function of moderate decrease on $\mathbb{R}$ whose Fourier transform $\hat{f}$ is continuous and satisfies

$$\hat{f}(\xi) = O\left(\frac{1}{|\xi|^{1+\alpha}}\right) \qquad \text{as } |\xi| \to \infty$$

for some $0 < \alpha < 1$. Prove that f satisfies a Hölder condition of order α, that is, that

$$|f(x+h) - f(x)| \leq M|h|^{\alpha} \qquad \text{for some } M > 0 \text{ and all } x, h \in \mathbb{R}.$$

(b) Let f be a continuous function on $\mathbb{R}$ which vanishes for $|x| \geq 1$, with $f(0) = 0$, and which is equal to $1/\log(1/|x|)$ for all x in a neighborhood of the origin. Prove that $\hat{f}$ is not of moderate decrease. In fact, there is no $\epsilon > 0$ so that $\hat{f}(\xi) = O(1/|\xi|^{1+\epsilon})$ as $|\xi| \to \infty$.

[Hint: For part (a), use the Fourier inversion formula to express $f(x+h) - f(x)$ as an integral involving $\hat{f}$, and estimate this integral separately for ξ in the two ranges $|\xi| \leq 1/|h|$ and $|\xi| \geq 1/|h|$.]

4. Bump functions. Examples of compactly supported functions in $\mathcal{S}(\mathbb{R})$ are very handy in many applications in analysis. Some examples are:

(a) Suppose $a < b$, and f is the function such that $f(x) = 0$ if $x \leq a$ or $x \geq b$ and

$$f(x) = e^{-1/(x-a)}e^{-1/(b-x)} \qquad \text{if } a < x < b.$$

Show that f is indefinitely differentiable on $\mathbb{R}$.

(b) Prove that there exists an indefinitely differentiable function F on $\mathbb{R}$ such that $F(x) = 0$ if $x \leq a$, $F(x) = 1$ if $x \geq b$, and F is strictly increasing on $[a, b]$.

(c) Let $\delta > 0$ be so small that $a + \delta < b - \delta$. Show that there exists an indefinitely differentiable function g such that g is 0 if $x \leq a$ or $x \geq b$, g is 1 on $[a + \delta, b - \delta]$, and g is strictly monotonic on $[a, a + \delta]$ and $[b - \delta, b]$.

[Hint: For (b) consider $F(x) = c\int_{-\infty}^{x} f(t)\,dt$ where c is an appropriate constant.]

5. Suppose f is continuous and of moderate decrease.

(a) Prove that $\hat{f}$ is continuous and $\hat{f}(\xi) \to 0$ as $|\xi| \to \infty$.

(b) Show that if $\hat{f}(\xi) = 0$ for all ξ, then f is identically 0.

[Hint: For part (a), show that $\hat{f}(\xi) = \frac{1}{2}\int_{-\infty}^{\infty}[f(x) - f(x - 1/(2\xi))]e^{-2\pi i x\xi}\,dx$. For part (b), verify that the multiplication formula $\int f(x)\hat{g}(x)\,dx = \int \hat{f}(y)g(y)\,dy$ still holds whenever $g \in \mathcal{S}(\mathbb{R})$.]

6. The function $e^{-\pi x^2}$ is its own Fourier transform. Generate other functions that (up to a constant multiple) are their own Fourier transforms. What must the constant multiples be? To decide this, prove that $\mathcal{F}^4 = I$. Here $\mathcal{F}(f) = \hat{f}$ is the Fourier transform, $\mathcal{F}^4 = \mathcal{F} \circ \mathcal{F} \circ \mathcal{F} \circ \mathcal{F}$, and I is the identity operator $(If)(x) = f(x)$ (see also Problem 7).

7. Prove that the convolution of two functions of moderate decrease is a function of moderate decrease.

[Hint: Write

$$\int f(x-y)g(y)\,dy = \int_{|y|\le|x|/2} + \int_{|y|\ge|x|/2}.$$

In the first integral $f(x-y) = O(1/(1+x^2))$ while in the second integral $g(y) = O(1/(1+x^2))$.]

8. Prove that f is continuous, of moderate decrease, and $\int_{-\infty}^{\infty} f(y)e^{-y^2}e^{2xy}dy = 0$ for all $x \in \mathbb{R}$, then $f = 0$.

[Hint: Consider $f * e^{-x^2}$.]

9. If f is of moderate decrease, then

$$\int_{-R}^{R} \left(1 - \frac{|\xi|}{R}\right) \hat{f}(\xi) e^{2\pi i x \xi} \, d\xi = (f * \mathcal{F}_R)(x), \tag{14}$$

where the Fejér kernel on the real line is defined by

$$\mathcal{F}_R(t) = \begin{cases} R\left(\dfrac{\sin \pi t R}{\pi t R}\right)^2 & \text{if } t \neq 0, \\ R & \text{if } t = 0. \end{cases}$$

Show that $\{\mathcal{F}_R\}$ is a family of good kernels as $R \to \infty$, and therefore (14) tends uniformly to $f(x)$ as $R \to \infty$. This is the analogue of Fejér's theorem for Fourier series in the context of the Fourier transform.

10. Below is an outline of a different proof of the Weierstrass approximation theorem.

Define the **Landau** kernels by

$$L_n(x) = \begin{cases} \dfrac{(1-x^2)^n}{c_n} & \text{if } -1 \le x \le 1, \\ 0 & \text{if } |x| \ge 1, \end{cases}$$

where c_n is chosen so that $\int_{-\infty}^{\infty} L_n(x)\,dx = 1$. Prove that $\{L_n\}_{n\ge 0}$ is a family of good kernels as $n \to \infty$. As a result, show that if f is a continuous function supported in $[-1/2, 1/2]$, then $(f * L_n)(x)$ is a sequence of polynomials on $[-1/2, 1/2]$ which converges uniformly to f.

[Hint: First show that $c_n \ge 2/(n+1)$.]

11. Suppose that u is the solution to the heat equation given by $u = f * \mathcal{H}_t$ where $f \in \mathcal{S}(\mathbb{R})$. If we also set $u(x,0) = f(x)$, prove that u is continuous on the closure of the upper half-plane, and vanishes at infinity, that is,

$$u(x,t) \to 0 \qquad \text{as } |x| + t \to \infty.$$

[Hint: To prove that u vanishes at infinity, show that (i) $|u(x,t)| \leq C/\sqrt{t}$ and (ii) $|u(x,t)| \leq C/(1+|x|^2) + Ct^{-1/2}e^{-cx^2/t}$. Use (i) when $|x| \leq t$, and (ii) otherwise.]

12. Show that the function defined by

$$u(x,t) = \frac{x}{t}\mathcal{H}_t(x)$$

satisfies the heat equation for $t > 0$ and $\lim_{t\to 0} u(x,t) = 0$ for every x, but u is *not* continuous at the origin.

[Hint: Approach the origin with (x,t) on the parabola $x^2/4t = c$ where c is a constant.]

13. Prove the following uniqueness theorem for harmonic functions in the strip $\{(x, y) : 0 < y < 1,\ -\infty < x < \infty\}$: if u is harmonic in the strip, continuous on its closure with $u(x, 0) = u(x, 1) = 0$ for all $x \in \mathbb{R}$, and u vanishes at infinity, then $u = 0$.

14. Prove that the periodization of the Fejér kernel $\mathcal{F}_N$ on the real line (Exercise 9) is equal to the Fejér kernel for periodic functions of period 1. In other words,

$$\sum_{n=-\infty}^{\infty} \mathcal{F}_N(x+n) = F_N(x),$$

when $N \geq 1$ is an integer, and where

$$F_N(x) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e^{2\pi i n x} = \frac{1}{N} \frac{\sin^2(N\pi x)}{\sin^2(\pi x)}.$$

15. This exercise provides another example of periodization.

(a) Apply the Poisson summation formula to the function g in Exercise 2 to obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)^2} = \frac{\pi^2}{(\sin \pi\alpha)^2}$$

whenever α is real, but not equal to an integer.

(b) Prove as a consequence that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\alpha)} = \frac{\pi}{\tan \pi\alpha} \tag{15}$$

whenever α is real but not equal to an integer. [Hint: First prove it when $0 < \alpha < 1$. To do so, integrate the formula in (b). What is the precise meaning of the series on the left-hand side of (15)? Evaluate at $\alpha = 1/2$.]

16. The Dirichlet kernel on the real line is defined by

$$\int_{-R}^{R} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi = (f * \mathcal{D}_R)(x) \qquad \text{so that} \qquad \mathcal{D}_R(x) = \widehat{\chi_{[-R,R]}}(x) = \frac{\sin(2\pi R x)}{\pi x}.$$

Also, the modified Dirichlet kernel for periodic functions of period 1 is defined by

$$D_N^*(x) = \sum_{|n|\leq N-1} e^{2\pi i n x} + \frac{1}{2}(e^{-2\pi i N x} + e^{2\pi i N x}).$$

Show that the result in Exercise 15 gives

$$\sum_{n=-\infty}^{\infty} \mathcal{D}_N(x+n) = D_N^*(x),$$

where $N \geq 1$ is an integer, and the infinite series must be summed symmetrically. In other words, the periodization of $\mathcal{D}_N$ is the modified Dirichlet kernel D_N^*.

17. The **gamma function** is defined for $s > 0$ by

$$\Gamma(s) = \int_0^\infty e^{-x} x^{s-1}\, dx.$$

(a) Show that for $s > 0$ the above integral makes sense, that is, that the following two limits exist:

$$\lim_{\substack{\delta \to 0 \\ \delta > 0}} \int_\delta^1 e^{-x} x^{s-1}\, dx \quad \text{and} \quad \lim_{A \to \infty} \int_1^A e^{-x} x^{s-1}\, dx.$$

(b) Prove that $\Gamma(s+1) = s\Gamma(s)$ whenever $s > 0$, and conclude that for every integer $n \geq 1$ we have $\Gamma(n+1) = n!$.

(c) Show that

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad \text{and} \quad \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

[Hint: For (c), use $\int_{-\infty}^{\infty} e^{-\pi x^2}\, dx = 1$.]

18. The **zeta function** is defined for $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s$. Verify the identity

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{2}\int_0^\infty t^{\frac{s}{2}-1}(\vartheta(t)-1)\,dt \qquad \text{whenever } s > 1$$

where Γ and ϑ are the gamma and theta functions, respectively:

$$\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}\,dt \qquad \text{and} \qquad \vartheta(s) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 s}.$$

More about the zeta function and its relation to the prime number theorem can be found in Book II.

19. The following is a variant of the calculation of $\zeta(2m) = \sum_{n=1}^{\infty} 1/n^{2m}$ found in Problem 4, Chapter 3.

(a) Apply the Poisson summation formula to $f(x) = t/(\pi(x^2+t^2))$ and $\hat{f}(\xi) = e^{-2\pi t|\xi|}$ where $t > 0$ in order to get

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \sum_{n=-\infty}^{\infty} e^{-2\pi t|n|}.$$

(b) Prove the following identity valid for $0 < t < 1$:

$$\frac{1}{\pi}\sum_{n=-\infty}^{\infty}\frac{t}{t^2+n^2} = \frac{1}{\pi t} + \frac{2}{\pi}\sum_{m=1}^{\infty}(-1)^{m+1}\zeta(2m)t^{2m-1}$$

as well as

$$\sum_{n=-\infty}^{\infty} e^{-2\pi t|n|} = \frac{2}{1-e^{-2\pi t}} - 1.$$

(c) Use the fact that

$$\frac{z}{e^z-1} = 1 - \frac{z}{2} + \sum_{m=1}^{\infty}\frac{B_{2m}}{(2m)!}z^{2m},$$

where B_k are the Bernoulli numbers to deduce from the above formula,

$$2\zeta(2m) = (-1)^{m+1}\frac{(2\pi)^{2m}}{(2m)!}B_{2m}.$$

20. The following results are relevant in information theory when one tries to recover a signal from its samples.

Suppose f is of moderate decrease and that its Fourier transform $\hat{f}$ is supported in $I = [-1/2, 1/2]$. Then, f is entirely determined by its restriction to $\mathbb{Z}$. This means that if g is another function of moderate decrease whose Fourier transform is supported in I and $f(n) = g(n)$ for all $n \in \mathbb{Z}$, then $f = g$. More precisely:

(a) Prove that the following reconstruction formula holds:

$$f(x) = \sum_{n=-\infty}^{\infty} f(n)K(x-n) \qquad \text{where } K(y) = \frac{\sin \pi y}{\pi y}.$$

Note that $K(y) = O(1/|y|)$ as $|y| \to \infty$.

(b) If $\lambda > 1$, then

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{1}{\lambda} f\left(\frac{n}{\lambda}\right) K_\lambda\left(x - \frac{n}{\lambda}\right) \qquad \text{where } K_\lambda(y) = \frac{\cos \pi y - \cos \pi \lambda y}{\pi^2(\lambda - 1)y^2}.$$

Thus, if one samples f "more often," the series in the reconstruction formula converges faster since $K_\lambda(y) = O(1/|y|^2)$ as $|y| \to \infty$. Note that $K_\lambda(y) \to K(y)$ as $\lambda \to 1$.

(c) Prove that $\displaystyle\int_{-\infty}^{\infty} |f(x)|^2\, dx = \sum_{n=-\infty}^{\infty} |f(n)|^2$.

[Hint: For part (a) show that if χ is the characteristic function of I, then $\hat{f}(\xi) = \chi(\xi) \sum_{n=-\infty}^{\infty} f(n)e^{-2\pi i n\xi}$. For (b) use the function in Figure 2 instead of $\chi(\xi)$.]

21. Suppose that f is continuous on $\mathbb{R}$. Show that f and $\hat{f}$ cannot both be compactly supported unless $f = 0$. This can be viewed in the same spirit as the uncertainty principle.

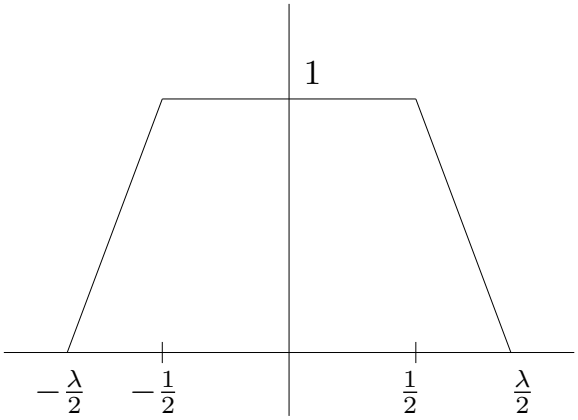


Figure 2. The function in Exercise 20

[Hint: Assume f is supported in $[0, 1/2]$. Expand f in a Fourier series in the interval $[0, 1]$, and note that as a result, f is a trigonometric polynomial.]

22. The heuristic assertion stated before Theorem 4.1 can be made precise as follows. If F is a function on $\mathbb{R}$, then we say that the preponderance of its mass is contained in an interval I (centered at the origin) if

$$\int_I x^2 |F(x)|^2 \, dx \geq \frac{1}{2} \int_{\mathbb{R}} x^2 |F(x)|^2 \, dx. \tag{16}$$

Now suppose $f \in \mathcal{S}$, and (16) holds with $F = f$ and $I = I_1$; also with $F = \hat{f}$ and $I = I_2$. Then if L_j denotes the length of I_j, we have

$$L_1 L_2 \geq \frac{1}{2\pi}.$$

A similar conclusion holds if the intervals are not necessarily centered at the origin.

23. The Heisenberg uncertainty principle can be formulated in terms of the operator $L = -\frac{d^2}{dx^2} + x^2$, which acts on Schwartz functions by the formula

$$L(f) = -\frac{d^2 f}{dx^2} + x^2 f.$$

This operator, sometimes called the **Hermite operator**, is the quantum analogue of the harmonic oscillator. Consider the usual inner product on $\mathcal{S}$ given by

$$(f, g) = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\, dx \qquad \text{whenever } f, g \in \mathcal{S}.$$

(a) Prove that the Heisenberg uncertainty principle implies

$$(Lf, f) \geq (f, f) \qquad \text{for all } f \in \mathcal{S}.$$

This is usually denoted by $L \geq I$. [Hint: Integrate by parts.]

(b) Consider the operators A and A^* defined on $\mathcal{S}$ by

$$A(f) = \frac{df}{dx} + xf \qquad \text{and} \qquad A^*(f) = -\frac{df}{dx} + xf.$$

The operators A and A^* are sometimes called the **annihilation** and **creation** operators, respectively. Prove that for all $f, g \in \mathcal{S}$ we have

(i) $(Af, g) = (f, A^* g)$,
(ii) $(Af, Af) = (A^* Af, f) \geq 0$,
(iii) $A^* A = L - I$.

In particular, this again shows that $L \geq I$.

(c) Now for $t \in \mathbb{R}$, let

$$A_t(f) = \frac{df}{dx} + txf \qquad \text{and} \qquad A_t^*(f) = -\frac{df}{dx} + txf.$$

Use the fact that $(A_t^* A_t f, f) \geq 0$ to give another proof of the Heisenberg uncertainty principle which says that whenever $\int_{-\infty}^{\infty} |f(x)|^2\, dx = 1$ then

$$\left(\int_{-\infty}^{\infty} x^2 |f(x)|^2\, dx\right)\left(\int_{-\infty}^{\infty} \left|\frac{df}{dx}\right|^2\, dx\right) \geq 1/4.$$

[Hint: Think of $(A_t^* A_t f, f)$ as a quadratic polynomial in t.]

Chapter 6. The Fourier Transform on $\mathbf{R}^d$

1. Suppose that R is a rotation in the plane $\mathbb{R}^2$, and let

$$R = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

denote its matrix with respect to the standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$.

(a) Write the conditions $R^t = R^{-1}$ and $\det(R) = \pm 1$ in terms of equations in a, b, c, d.

(b) Show that there exists $\varphi \in \mathbb{R}$ such that $a + ib = e^{i\varphi}$.

(c) Conclude that if R is proper, then it can be expressed as $z \mapsto ze^{i\varphi}$, and if R is improper, then it takes the form $z \mapsto \overline{z}e^{i\varphi}$, where $\overline{z} = x - iy$.

2. Suppose that $R : \mathbb{R}^3 \to \mathbb{R}^3$ is a proper rotation.

(a) Show that $p(t) = \det(R - tI)$ is a polynomial of degree 3, and prove that there exists $\gamma \in S^2$ (where S^2 denotes the unit sphere in $\mathbb{R}^3$) with

$$R(\gamma) = \gamma.$$

[Hint: Use the fact that $p(0) > 0$ to see that there is $\lambda > 0$ with $p(\lambda) = 0$. Then $R - \lambda I$ is singular, so its kernel is non-trivial.]

(b) If $\mathcal{P}$ denotes the plane perpendicular to γ and passing through the origin, show that

$$R : \mathcal{P} \to \mathcal{P},$$

and that this linear map is a rotation.

3. Recall the formula

$$\int_{\mathbb{R}^d} F(x)\,dx = \int_{S^{d-1}} \int_0^\infty F(r\gamma) r^{d-1}\,dr\,d\sigma(\gamma).$$

Apply this to the special case when $F(x) = g(r)f(\gamma)$, where $x = r\gamma$, to prove that for any rotation R, one has

$$\int_{S^{d-1}} f(R(\gamma))\,d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\,d\sigma(\gamma),$$

whenever f is a continuous function on the sphere S^{d-1}.

4. Let A_d and V_d denote the area and volume of the unit sphere and unit ball in $\mathbb{R}^d$, respectively.

(a) Prove the formula

$$A_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$$

so that $A_2 = 2\pi$, $A_3 = 4\pi$, $A_4 = 2\pi^2, \ldots$. Here $\Gamma(x) = \int_0^\infty e^{-t} t^{x-1}\, dt$ is the Gamma function. [Hint: Use polar coordinates and the fact that $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\, dx = 1$.]

(b) Show that $d\, V_d = A_d$, hence

$$V_d = \frac{\pi^{d/2}}{\Gamma(d/2+1)}.$$

In particular $V_2 = \pi$, $V_3 = 4\pi/3, \ldots$.

5. Let A be a $d \times d$ positive definite symmetric matrix with real coefficients. Show that

$$\int_{\mathbb{R}^d} e^{-\pi(x,A(x))}\, dx = (\det(A))^{-1/2}.$$

This generalizes the fact that $\int_{\mathbb{R}^d} e^{-\pi|x|^2}\, dx = 1$, which corresponds to the case where A is the identity.

[Hint: Apply the spectral theorem to write $A = RDR^{-1}$ where R is a rotation and, D is diagonal with entries $\lambda_1, \ldots, \lambda_d$, where $\{\lambda_i\}$ are the eigenvalues of A.]

6. Suppose $\psi \in \mathcal{S}(\mathbb{R}^d)$ satisfies $\int |\psi(x)|^2\,dx = 1$. Show that

$$\left(\int_{\mathbb{R}^d} |x|^2 |\psi(x)|^2\,dx\right)\left(\int_{\mathbb{R}^d} |\xi|^2 |\hat{\psi}(\xi)|^2\,d\xi\right) \geq \frac{d^2}{16\pi^2}.$$

This is the statement of the Heisenberg uncertainty principle in d dimensions.

7. Consider the time-dependent heat equation in $\mathbb{R}^d$:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2}, \qquad \text{where } t > 0, \tag{15}$$

with boundary values $u(x,0) = f(x) \in \mathcal{S}(\mathbb{R}^d)$. If

$$\mathcal{H}_t^{(d)}(x) = \frac{1}{(4\pi t)^{d/2}} e^{-|x|^2/4t} = \int_{\mathbb{R}^d} e^{-4\pi^2 t|\xi|^2} e^{2\pi i x\cdot\xi}\, d\xi$$

is the d-dimensional **heat kernel**, show that the convolution

$$u(x,t) = (f * \mathcal{H}_t^{(d)})(x)$$

is indefinitely differentiable when $x \in \mathbb{R}^d$ and $t > 0$. Moreover, u solves (15), and is continuous up to the boundary $t = 0$ with $u(x,0) = f(x)$.

The reader may also wish to formulate the d-dimensional analogues of Theorem 2.1 and 2.3 in Chapter 5.

8. In Chapter 5, we found that a solution to the steady-state heat equation in the upper half-plane with boundary values f is given by the convolution $u = f * \mathcal{P}_y$ where the Poisson kernel is

$$\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2} \qquad \text{where } x \in \mathbb{R} \text{ and } y > 0.$$

More generally, one can calculate the d-dimensional Poisson kernel using the Fourier transform as follows.

(a) The **subordination principle** allows one to write expressions involving the function e^{-x} in terms of corresponding expressions involving the function e^{-x^2}. One form of this is the identity

$$e^{-\beta} = \int_0^\infty \frac{e^{-u}}{\sqrt{\pi u}} e^{-\beta^2/4u}\, du$$

when $\beta \geq 0$. Prove this identity with $\beta = 2\pi|x|$ by taking the Fourier transform of both sides.

(b) Consider the steady-state heat equation in the upper half-space $\{(x, y) : x \in \mathbb{R}^d,\ y > 0\}$

$$\sum_{j=1}^{d} \frac{\partial^2 u}{\partial x_j^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

with the Dirichlet boundary condition $u(x, 0) = f(x)$. A solution to this problem is given by the convolution $u(x, y) = (f * P_y^{(d)})(x)$ where $P_y^{(d)}(x)$ is the d-dimensional Poisson kernel

$$P_y^{(d)}(x) = \int_{\mathbb{R}^d} e^{2\pi i x\cdot\xi} e^{-2\pi|\xi|y}\, d\xi.$$

Compute $P_y^{(d)}(x)$ by using the subordination principle and the d-dimensional heat kernel. (See Exercise 7.) Show that

$$P_y^{(d)}(x) = \frac{\Gamma((d+1)/2)}{\pi^{(d+1)/2}} \frac{y}{(|x|^2+y^2)^{(d+1)/2}}.$$

9. A **spherical wave** is a solution $u(x,t)$ of the Cauchy problem for the wave equation in $\mathbb{R}^d$, which as a function of x is radial. Prove that u is a spherical wave if and only if the initial data $f, g \in \mathcal{S}$ are both radial.

10. Let $u(x,t)$ be a solution of the wave equation, and let $E(t)$ denote the energy of this wave

$$E(t) = \int_{\mathbb{R}^d} \left| \frac{\partial u}{\partial t}(x,t) \right|^2 + \sum_{j=1}^{d} \int_{\mathbb{R}^d} \left| \frac{\partial u}{\partial x_j}(x,t) \right|^2 \, dx.$$

We have seen that $E(t)$ is constant using Plancherel's formula. Give an alternate proof of this fact by differentiating the integral with respect to t and showing that

$$\frac{dE}{dt} = 0.$$

[Hint: Integrate by parts.]

11. Show that the solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2}$$

subject to $u(x,0) = f(x)$ and $\frac{\partial u}{\partial t}(x,0) = g(x)$, where $f, g \in \mathcal{S}(\mathbb{R}^3)$, is given by

$$u(x,t) = \frac{1}{|S(x,t)|} \int_{S(x,t)} [tg(y) + f(y) + \nabla f(y) \cdot (y-x)]\, d\sigma(y),$$

where $S(x,t)$ denotes the sphere of center x and radius t, and $|S(x,t)|$ its area. This is an alternate expression for the solution of the wave equation given in Theorem 3.6. It is sometimes called **Kirchhoff's** formula.

12. Establish the identity (14) about the dual transform given in the text. In other words, prove that

$$\int_{\mathbb{R}} \int_{S^2} \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)} d\sigma(\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\overline{\mathcal{R}^*(F)(x)}\, dx \tag{16}$$

where $f \in \mathcal{S}(\mathbb{R}^3)$, $F \in \mathcal{S}(\mathbb{R} \times S^2)$, and

$$\mathcal{R}(f) = \int_{\mathcal{P}_{t,\gamma}} f \quad \text{and} \quad \mathcal{R}^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma)\, d\sigma(\gamma).$$

[Hint: Consider the integral

$$\int\int\int f(t\gamma + u_1 e_2 + u_2 e_2)\overline{F(t,\gamma)}\, dt\, d\sigma(\gamma)\, du_1\, du_2.$$

Integrating first in u gives the left-hand side of (16), while integrating in u and t and setting $x = t\gamma + u_1 e_2 + u_2 e_2$ gives the right-hand side.]

13. For each (t, θ) with $t \in \mathbb{R}$ and $|\theta| \leq \pi$, let $L = L_{t,\theta}$ denote the line in the (x, y)-plane given by

$$x \cos\theta + y \sin\theta = t.$$

This is the line perpendicular to the direction $(\cos\theta, \sin\theta)$ at "distance" t from the origin (we allow negative t). For $f \in \mathcal{S}(\mathbb{R}^2)$ the X-ray transform or two-dimensional Radon transform of f is defined by

$$X(f)(t, \theta) = \int_{L_{t,\theta}} f = \int_{-\infty}^{\infty} f(t\cos\theta + u\sin\theta, t\sin\theta - u\cos\theta)\, du.$$

Calculate the X-ray transform of the function $f(x, y) = e^{-\pi(x^2+y^2)}$.

14. Let X be the X-ray transform. Show that if $f \in \mathcal{S}$ and $X(f) = 0$, then $f = 0$, by taking the Fourier transform in one variable.

15. For $F \in \mathcal{S}(\mathbb{R} \times S^1)$, define the **dual X-ray transform** $X^*(F)$ by integrating F over all lines that pass through the point (x, y) (that is, those lines $L_{t,\theta}$ with $x\cos\theta + y\sin\theta = t$):

$$X^*(F)(x, y) = \int F(x\cos\theta + y\sin\theta, \theta)\, d\theta.$$

Check that in this case, if $f \in \mathcal{S}(\mathbb{R}^2)$ and $F \in \mathcal{S}(\mathbb{R} \times S^1)$, then

$$\int\int X(f)(t, \theta)\overline{F(t, \theta)}\, dt\, d\theta = \int\int f(x, y)\overline{X^*(F)(x, y)}\, dx\, dy.$$

Chapter 7. Finite Fourier Analysis

1. Let f be a function on the circle. For each $N \geq 1$ the discrete Fourier coefficients of f are defined by

$$a_N(n) = \frac{1}{N} \sum_{k=1}^{N} f(e^{2\pi ik/N}) e^{-2\pi ikn/N}, \qquad \text{for } n \in \mathbb{Z}.$$

We also let

$$a(n) = \int_0^1 f(e^{2\pi ix}) e^{-2\pi inx}\, dx$$

denote the ordinary Fourier coefficients of f.

(a) Show that $a_N(n) = a_N(n+N)$.

(b) Prove that if f is continuous, then $a_N(n) \to a(n)$ as $N \to \infty$.

2. If f is a C^1 function on the circle, prove that $|a_N(n)| \le c/|n|$ whenever $0 < |n| \le N/2$.

[Hint: Write

$$a_N(n)[1 - e^{2\pi i \ell n/N}] = \frac{1}{N} \sum_{k=1}^{N} [f(e^{2\pi i k/N}) - f(e^{2\pi i (k+\ell)/N})] e^{-2\pi i k n/N},$$

and choose ℓ so that $\ell n/N$ is nearly $1/2$.]

3. By a similar method, show that if f is a C^2 function on the circle, then

$$|a_N(n)| \le c/|n|^2, \qquad \text{whenever } 0 < |n| \le N/2.$$

As a result, prove the inversion formula for $f \in C^2$,

$$f(e^{2\pi ix}) = \sum_{n=-\infty}^{\infty} a(n) e^{2\pi inx}$$

from its finite version.
[Hint: For the first part, use the second symmetric difference

$$f(e^{2\pi i(k+\ell)/N}) + f(e^{2\pi i(k-\ell)/N}) - 2f(e^{2\pi ik/N}).$$

For the second part, if N is odd (say), write the inversion formula as

$$f(e^{2\pi ik/N}) = \sum_{|n|<N/2} a_N(n) e^{2\pi ikn/N}.]$$

4. Let e be a character on $G = \mathbb{Z}(N)$, the additive group of integers modulo N. Show that there exists a unique $0 \leq \ell \leq N-1$ so that

$$e(k) = e_\ell(k) = e^{2\pi i \ell k/N} \qquad \text{for all } k \in \mathbb{Z}(N).$$

Conversely, every function of this type is a character on $\mathbb{Z}(N)$. Deduce that $e_\ell \mapsto \ell$ defines an isomorphism from $\hat{G}$ to G.

[Hint: Show that $e(1)$ is an Nth root of unity.]

5. Show that all characters on S^1 are given by

$$e_n(x) = e^{2\pi i n x} \qquad \text{with } n \in \mathbb{Z},$$

and check that $e_n \mapsto n$ defines an isomorphism from $\widehat{S^1}$ to $\mathbb{Z}$.
[Hint: If F is continuous and $F(x+y) = F(x)F(y)$, then F is differentiable. To see this, note that if $F(0) \neq 0$, then for appropriate δ, $c = \int_0^\delta F(y)\,dy \neq 0$, and $cF(x) = \int_x^{\delta+x} F(y)\,dy$. Differentiate to conclude that $F(x) = e^{Ax}$ for some A.]

6. Prove that all characters on $\mathbb{R}$ take the form

$$e_\xi(x) = e^{2\pi i \xi x} \qquad \text{with } \xi \in \mathbb{R},$$

and that $e_\xi \mapsto \xi$ defines an isomorphism from $\widehat{\mathbb{R}}$ to $\mathbb{R}$. The argument in Exercise 5 applies here as well.

7. Let $\zeta = e^{2\pi i/N}$. Define the $N \times N$ matrix $M = (a_{jk})_{1 \le j,k \le N}$ by $a_{jk} = N^{-1/2}\zeta^{jk}$.

(a) Show that M is unitary.

(b) Interpret the identity $(Mu, Mv) = (u, v)$ and the fact that $M^* = M^{-1}$ in terms of Fourier series on $\mathbb{Z}(N)$.

8. Suppose that $P(x) = \sum_{n=1}^{N} a_n e^{2\pi i n x}$.

(a) Show by using the Parseval identities for the circle and $\mathbb{Z}(N)$, that

$$\int_0^1 |P(x)|^2 \, dx = \frac{1}{N} \sum_{j=1}^{N} |P(j/N)|^2.$$

(b) Prove the reconstruction formula

$$P(x) = \sum_{j=1}^{N} P(j/N) K(x - (j/N))$$

where

$$K(x) = \frac{e^{2\pi i x}}{N} \frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} = \frac{1}{N}(e^{2\pi i x} + e^{2\pi i 2x} + \cdots + e^{2\pi i N x}).$$

Observe that P is completely determined by the values $P(j/N)$ for $1 \le j \le N$. Note also that $K(0) = 1$, and $K(j/N) = 0$ whenever j is not congruent to 0 modulo N.

9. To prove the following assertions, modify the argument given in the text.

(a) Show that one can compute the Fourier coefficients of a function on $\mathbb{Z}(N)$ when $N = 3^n$ with at most $6N \log_3 N$ operations.

(b) Generalize this to $N = \alpha^n$ where α is an integer > 1.

10. A group G is **cyclic** if there exists $g \in G$ that generates all of G, that is, if any element in G can be written as g^n for some $n \in \mathbb{Z}$. Prove that a finite abelian group is cyclic if and only if it is isomorphic to $\mathbb{Z}(N)$ for some N.

11. Write down the multiplicative tables for the groups $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, $\mathbb{Z}^*(6)$, $\mathbb{Z}^*(8)$, and $\mathbb{Z}^*(9)$. Which of these groups are cyclic?

12. Suppose that G is a finite abelian group and $e : G \to \mathbb{C}$ is a function that satisfies $e(x \cdot y) = e(x)e(y)$ for all $x, y \in G$. Prove that either e is identically 0, or e never vanishes. In the second case, show that for each x, $e(x) = e^{2\pi i r}$ for some $r \in \mathbb{Q}$ of the form $r = p/q$, where $q = |G|$.

13. In analogy with ordinary Fourier series, one may interpret finite Fourier expansions using convolutions as follows. Suppose G is a finite abelian group, 1_G its unit, and V the vector space of complex-valued functions on G.

(a) The convolution of two functions f and g in V is defined for each $a \in G$ by

$$(f * g)(a) = \frac{1}{|G|} \sum_{b \in G} f(b) g(a \cdot b^{-1}).$$

Show that for all $e \in \hat{G}$ one has $\widehat{(f * g)}(e) = \hat{f}(e)\hat{g}(e)$.

(b) Use Theorem 2.5 to show that if e is a character on G, then

$$\sum_{e \in \hat{G}} e(c) = 0 \qquad \text{whenever } c \in G \text{ and } c \neq 1_G.$$

(c) As a result of (b), show that the Fourier series $Sf(a) = \sum_{e \in \hat{G}} \hat{f}(e)e(a)$ of a function $f \in V$ takes the form

$$Sf = f * D,$$

where D is defined by

$$D(c) = \sum_{e \in \hat{G}} e(c) = \begin{cases} |G| & \text{if } c = 1_G, \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

Since $f * D = f$, we recover the fact that $Sf = f$. Loosely speaking, D corresponds to a "Dirac delta function"; it has unit mass

$$\frac{1}{|G|} \sum_{c \in G} D(c) = 1,$$

and (4) says that this mass is concentrated at the unit element in G. Thus D has the same interpretation as the "limit" of a family of good kernels. (See Section 4, Chapter 2.)

Note. The function D reappears in the next chapter as $\delta_1(n)$.

Chapter 8. Dirichlet's Theorem

1. Prove that there are infinitely many primes by observing that if there were only finitely many, $p_1, \ldots, p_N$, then

$$\prod_{j=1}^{N} \frac{1}{1-1/p_j} \geq \sum_{n=1}^{\infty} \frac{1}{n}.$$

2. In the text we showed that there are infinitely many primes of the form $4k + 3$ by a modification of Euclid's original argument. Adapt this technique to prove the similar result for primes of the form $3k + 2$, and for those of the form $6k + 5$.

3. Prove that if p and q are relatively prime, then $\mathbb{Z}^*(p) \times \mathbb{Z}^*(q)$ is isomorphic to $\mathbb{Z}^*(pq)$.

4. Let $\varphi(n)$ denote the number of positive integers $\leq n$ that are relatively prime to n. Use the previous exercise to show that if n and m are relatively prime, then

$$\varphi(nm) = \varphi(n)\varphi(m).$$

One can give a formula for the Euler phi-function as follows:

(a) Calculate $\varphi(p)$ when p is prime by counting the number of elements in $\mathbb{Z}^*(p)$.

(b) Give a formula for $\varphi(p^k)$ when p is prime and $k \geq 1$ by counting the number of elements in $\mathbb{Z}^*(p^k)$.

(c) Show that

$$\varphi(n) = n \prod_i \left(1 - \frac{1}{p_i}\right),$$

where p_i are the primes that divide n.

5. If n is a positive integer, show that

$$n = \sum_{d|n} \varphi(d),$$

where φ is the Euler phi-function.
[Hint: There are precisely $\varphi(n/d)$ integers $1 \leq m \leq n$ with $\gcd(m, n) = d$.]

6. Write down the characters of the groups $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, $\mathbb{Z}^*(6)$, and $\mathbb{Z}^*(8)$.

(a) Which ones are real, or complex?

(b) Which ones are even, or odd? (A character is even if $\chi(-1) = 1$, and odd otherwise).

7. Recall that for $|z| < 1$,

$$\log_1\left(\frac{1}{1-z}\right) = \sum_{k\geq 1} \frac{z^k}{k}.$$

We have seen that

$$e^{\log_1\left(\frac{1}{1-z}\right)} = \frac{1}{1-z}.$$

(a) Show that if $w = 1/(1-z)$, then $|z| < 1$ if and only if $\mathrm{Re}(w) > 1/2$.

(b) Show that if $\mathrm{Re}(w) > 1/2$ and $w = \rho e^{i\varphi}$ with $\rho > 0$, $|\varphi| < \pi$, then

$$\log_1 w = \log \rho + i\varphi.$$

[Hint: If $e^{\zeta} = w$, then the real part of ζ is uniquely determined and its imaginary part is determined modulo 2π.]

8. Let ζ denote the zeta function defined for $s > 1$.

(a) Compare $\zeta(s)$ with $\int_1^\infty x^{-s}\,dx$ to show that

$$\zeta(s) = \frac{1}{s-1} + O(1) \qquad \text{as } s \to 1^+.$$

(b) Prove as a consequence that

$$\sum_p \frac{1}{p^s} = \log\left(\frac{1}{s-1}\right) + O(1) \qquad \text{as } s \to 1^+.$$

9. Let χ_0 denote the trivial Dirichlet character mod q, and $p_1, \ldots, p_k$ the distinct prime divisors of q. Recall that $L(s, \chi_0) = (1 - p_1^{-s}) \cdots (1 - p_k^{-s})\zeta(s)$, and show as a consequence

$$L(s, \chi_0) = \frac{\varphi(q)}{q} \frac{1}{s-1} + O(1) \qquad \text{as } s \to 1^+.$$

[Hint: Use the asymptotics for ζ in Exercise 8.]

10. Show that if ℓ is relatively prime to q, then

$$\sum_{p \equiv \ell} \frac{1}{p^s} = \frac{1}{\varphi(q)} \log\left(\frac{1}{s-1}\right) + O(1) \qquad \text{as } s \to 1^+.$$

This is a quantitative version of Dirichlet's theorem.
[Hint: Recall (4).]

11. Use the characters for $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$, $\mathbb{Z}^*(5)$, and $\mathbb{Z}^*(6)$ to verify directly that $L(1,\chi) \neq 0$ for all non-trivial Dirichlet characters modulo q when $q = 3, 4, 5,$ and 6.

[Hint: Consider in each case the appropriate alternating series.]

12. Suppose χ is real and non-trivial; assuming the theorem that $L(1,\chi) \neq 0$, show directly that $L(1,\chi) > 0$.

[Hint: Use the product formula for $L(s,\chi)$.]

13. Let $\{a_n\}_{n=-\infty}^{\infty}$ be a sequence of complex numbers such that $a_n = a_m$ if $n = m \bmod q$. Show that the series

$$\sum_{n=1}^{\infty} \frac{a_n}{n}$$

converges if and only if $\sum_{n=1}^{q} a_n = 0$.

[Hint: Sum by parts.]

14. The series

$$F(\theta) = \sum_{|n|\neq 0} \frac{e^{in\theta}}{n}, \qquad \text{for } |\theta| < \pi,$$

converges for every θ and is the Fourier series of the function defined on $[-\pi, \pi]$ by $F(0) = 0$ and

$$F(\theta) = \begin{cases} i(-\pi - \theta) & \text{if } -\pi \leq \theta < 0 \\ i(\pi - \theta) & \text{if } 0 < \theta \leq \pi, \end{cases}$$

and extended by periodicity (period 2π) to all of $\mathbb{R}$ (see Exercise 8 in Chapter 2).

Show also that if $\theta \neq 0 \bmod 2\pi$, then the series

$$E(\theta) = \sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$$

converges, and that

$$E(\theta) = \frac{1}{2} \log \left(\frac{1}{2 - 2\cos\theta} \right) + \frac{i}{2} F(\theta).$$

15. To sum the series $\sum_{n=1}^{\infty} a_n/n$, with $a_n = a_m$ if $n = m \bmod q$ and $\sum_{n=1}^{q} a_n = 0$, proceed as follows.

(a) Define

$$A(m) = \sum_{n=1}^{q} a_n \zeta^{-mn} \qquad \text{where } \zeta = e^{2\pi i/q}.$$

Note that $A(q) = 0$. With the notation of the previous exercise, prove that

$$\sum_{n=1}^{\infty} \frac{a_n}{n} = \frac{1}{q} \sum_{m=1}^{q-1} A(m) E(2\pi m/q).$$

[Hint: Use Fourier inversion on $\mathbb{Z}(q)$.]

(b) If $\{a_m\}$ is odd, $(a_{-m} = -a_m)$ for $m \in \mathbb{Z}$, observe that $a_0 = a_q = 0$ and show that

$$A(m) = \sum_{1 \le n < q/2} a_n (\zeta^{-mn} - \zeta^{mn}).$$

(c) Still assuming that $\{a_m\}$ is odd, show that

$$\sum_{n=1}^{\infty} \frac{a_n}{n} = \frac{1}{2q} \sum_{m=1}^{q-1} A(m) F(2\pi m/q).$$

[Hint: Define $\tilde{A}(m) = \sum_{n=1}^{q} a_n \zeta^{mn}$ and apply the Fourier inversion formula.]

16. Use the previous exercises to show that

$$\frac{\pi}{3\sqrt{3}} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{5} + \frac{1}{7} - \frac{1}{8} + \cdots,$$

which is $L(1, \chi)$ for the non-trivial (odd) Dirichlet character modulo 3.